SLIIT UNI
THE KNOWLEDGE UNIVERSITY

Sri Lanka Institute of Information Technology

Faculty of Computing

Complex Numbers

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Outline

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2. The symbol j
3. Powers of j
4. Complex numbers
5. Graphical representation of a complex number
6. Polar form of a complex number
7. Exponential form of a complex number

Introduction

Ideas and symbols

- The numerals were devised to enable written calculations and records of quantities and measurements. When a grouping of symbols such as $\sqrt{-1}$ occurs to which there is no corresponding quantity we ask ourselves why such a grouping occurs and can we make anything of it?
- In response we carry on manipulating with it to see if anything worthwhile comes to light.
- We call $\sqrt{-1}$ an **imaginary** number to distinguish it from those numbers to which we can associate quantity which we call **real** numbers.

The symbol j

Quadratic equations

- The solutions to the quadratic equation:

$$x^2 - 1 = 0$$

are:

$$x = +1 \text{ and } x = -1$$

- The solutions to the quadratic equation:

$$x^2 + 1 = 0$$

are:

$$x = +\sqrt{-1} \text{ and } x = -\sqrt{-1}$$

We avoid the clumsy notation by defining $j = \sqrt{-1}$

Powers of j

- Positive integer powers
- Negative integer powers

Powers of j

Positive integer powers

Because: $j = \sqrt{-1}$

so:

$$j^2 = -1$$

$$j^3 = j^2 j = -j$$

$$j^4 = \left(j^2\right)^2 = (-1)^2 = 1$$

$$j^5 = j^4 j = j$$

Powers of j

Negative integer powers

Because: $j^2 = -1$ so $j = -\frac{1}{j} = -j^{-1}$

and so: $j^{-1} = -j$

$$j^{-2} = \left(j^2\right)^{-1} = (-1)^{-1} = -1$$

$$j^{-3} = \left(j^{-2}\right)\left(j^{-1}\right) = (-1)(-j) = j$$

$$j^{-4} = \left(j^{-2}\right)^2 = (-1)^2 = 1$$

Complex numbers

Introduction

- A complex number is a mixture of a real number and an imaginary number. The symbol z is used to denote a complex number.
- In the complex number $z = 3 + j5$:
  - the number 3 is called the real part of z and denoted by Re(z)
  - the number 5 is called the imaginary part of z, denoted by Im(z)

Complex numbers

Addition and subtraction

- The real parts and the imaginary parts are added (subtracted) separately:
- and so:

$$
\begin{aligned}
&(4+j5)+(3-j2) \\
&\qquad = 4+j5+3-j2 \\
&\qquad = 4+3+j5-j2 \\
&\qquad = 7+j3
\end{aligned}
$$

Complex numbers

Multiplication

- Complex numbers are multiplied just like any other binomial product:
- and so:

$$
\begin{aligned}
&(4+j5)\times(3-j2)\\
&\qquad = 4(3-j2)+j5(3-j2)\\
&\qquad = 12-j8+j15-j^2 10\\
&\qquad = 12-j8+j15+10 \quad \text{because } j^2=-1\\
&\qquad = 22+j7
\end{aligned}
$$

Complex numbers

Complex conjugate

- The complex conjugate of a complex number is obtained by switching the sign of the imaginary part. So that:

$$(5 + j8) \text{ and } (5 - j8)$$

- Are complex conjugates of each other.
- The product of a complex number and its complex conjugate is entirely real:

$$\begin{aligned}
&(a + jb) \times (a - jb) \\
&\qquad = a(a - jb) + jb(a - jb) \\
&\qquad = a^2 - jba + jba - j^2b^2 \\
&\qquad = a^2 + b^2
\end{aligned}$$

Complex numbers

Division

- To divide two complex numbers both numerator and denominator are multiplied by the complex conjugate of the denominator:

$$\begin{aligned}\frac{7-j4}{4+j3} &= \frac{(7-j4)}{(4+j3)} \times \frac{(4-j3)}{(4-j3)} \\ &= \frac{(7-j4)\times(4-j3)}{(4+j3)\times(4-j3)} \\ &= \frac{(16-j37)}{(16+9)} \\ &= \frac{16}{25} - j\frac{37}{25}\end{aligned}$$

Complex numbers

Equal complex numbers

- If two complex numbers are equal then their respective real parts are equal and their respective imaginary parts are equal.

$$\text{If } a + jb = c + jd \text{ then } a = c \text{ and } b = d$$

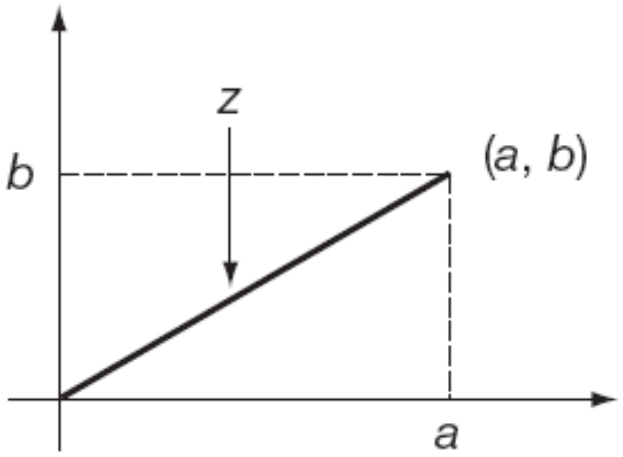
Graphical representation of a complex number

- Argand diagram
- Graphical addition of a complex number

Graphical representation of a complex number

Argand diagram

- The complex number $z = a + jb$ can be represented by the line joining the origin to the point (a, b) set against Cartesian axes.



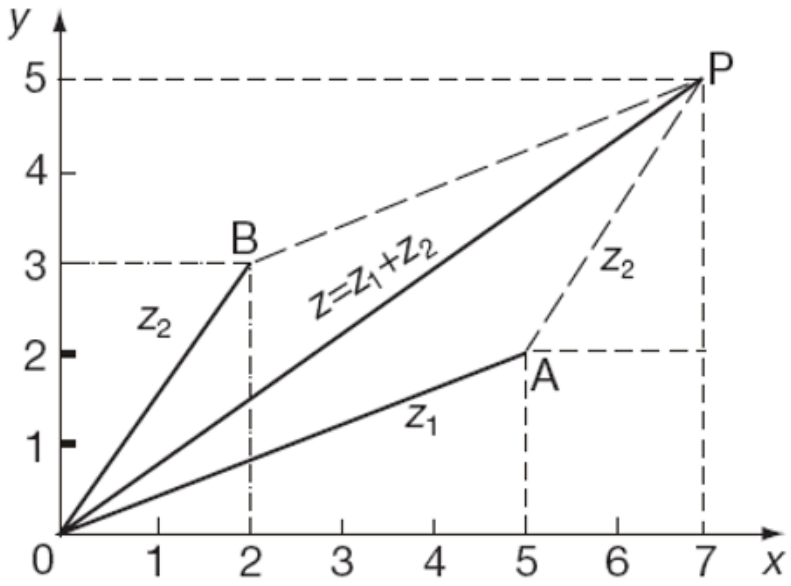
- This is called the **Argand** diagram and the plane of points is called the complex plane.

Graphical representation of a complex number

Graphical addition of complex numbers

- Complex numbers add (subtract) according to the parallelogram rule:

$$(5+j2)+(2+j3)=7+j5$$



Polar form of a complex number

A complex number can be expressed in polar coordinates r and θ

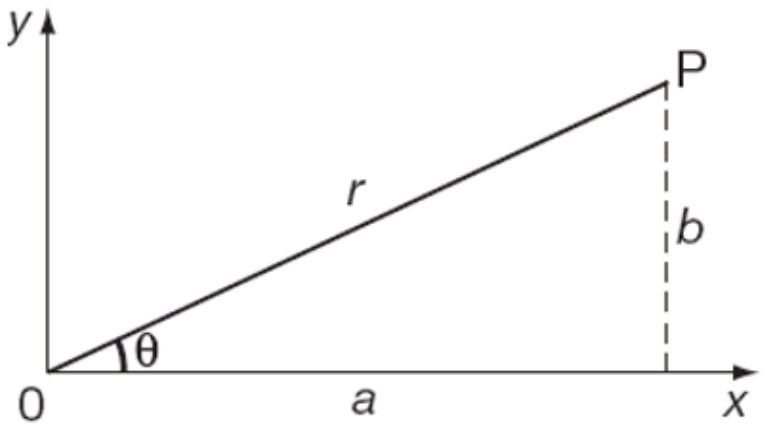
$$z = a + jb$$
$$= r(\cos\theta + j\sin\theta)$$

where:

$$a = r\cos\theta, \quad b = r\sin\theta$$

and:

$$r^2 = a^2 + b^2$$



Exponential form of a complex number

- Therefore:

$$\boxed{z = r\left(\cos\theta + j\sin\theta\right) = re^{j\theta}}$$

Polar-form calculations

Notation

- The polar form of a complex number is readily obtained from the Argand diagram of the number in Cartesian form.

Given:

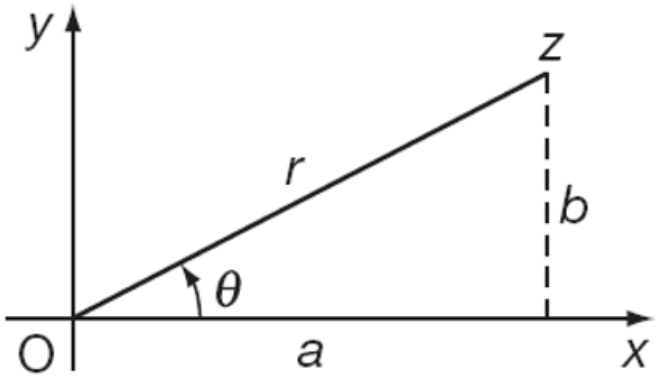
$$z = a + jb$$

then:

$$r^2 = a^2 + b^2 \quad \text{so} \quad r = \sqrt{a^2 + b^2}$$

and

$$\tan\theta = \frac{b}{a} \quad \text{so} \quad \theta = \tan^{-1}\frac{b}{a}$$



- The length r is called the **modulus** of the complex number and the angle θ is called the **argument** of the complex number

Polar-form calculations

Multiplication

- When two complex numbers, written in polar form, are multiplied the product is given as a complex number whose modulus is the product of the two moduli and whose argument is the sum of the two arguments.

$$\text{If } z_1 = r_1\left(\cos\theta_1 + j\sin\theta_1\right) \text{ and } z_2 = r_2\left(\cos\theta_2 + j\sin\theta_2\right)$$

$$\text{then } z_1 z_2 = r_1 r_2\left(\cos\left[\theta_1+\theta_2\right] + j\sin\left[\theta_1+\theta_2\right]\right)$$

Polar-form calculations

Division

- When two complex numbers, written in polar form, are divided the quotient is given as a complex number whose modulus is the quotient of the two moduli and whose argument is the difference of the two arguments.

$$\text{If } z_1 = r_1\left(\cos\theta_1 + j\sin\theta_1\right) \text{ and } z_2 = r_2\left(\cos\theta_2 + j\sin\theta_2\right)$$

$$\text{then } \frac{z_1}{z_2} = \frac{r_1}{r_2}\left(\cos\left[\theta_1 - \theta_2\right] + j\sin\left[\theta_1 - \theta_2\right]\right)$$

Roots of a complex number

- De Moivre's theorem
- nth roots

Roots of a complex number

De Moivre's theorem

- If a complex number is raised to the power n the result is a complex number whose modulus is the original modulus raised to the power n and whose argument is the original argument multiplied by n.

$$\text{If } z = r\left(\cos\theta + j\sin\theta\right)$$

$$\text{then } z^n = \left[r\left(\cos\theta + j\sin\theta\right)\right]^n = r^n\left(\cos n\theta + j\sin n\theta\right)$$

Thank You!